

Q4 Production Efficiency Report (ID: doc-q4-eff)

1. Executive Summary & Production Efficiency Analysis

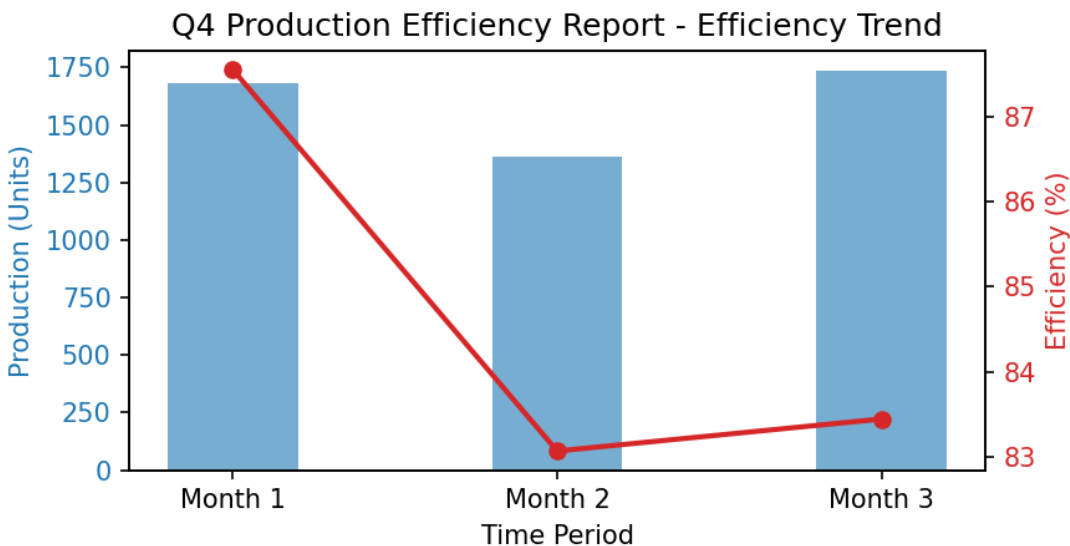
During the operational assessment of Q4 Production Efficiency Report, production efficiency experienced variance. The primary root cause for downtime in Q2 was unexpected hydraulic failure in Pump-04, whereas in Q4 the downtime was driven by supply chain shortages of raw titanium alloy.

2. Manufacturing Metrics & Output Table

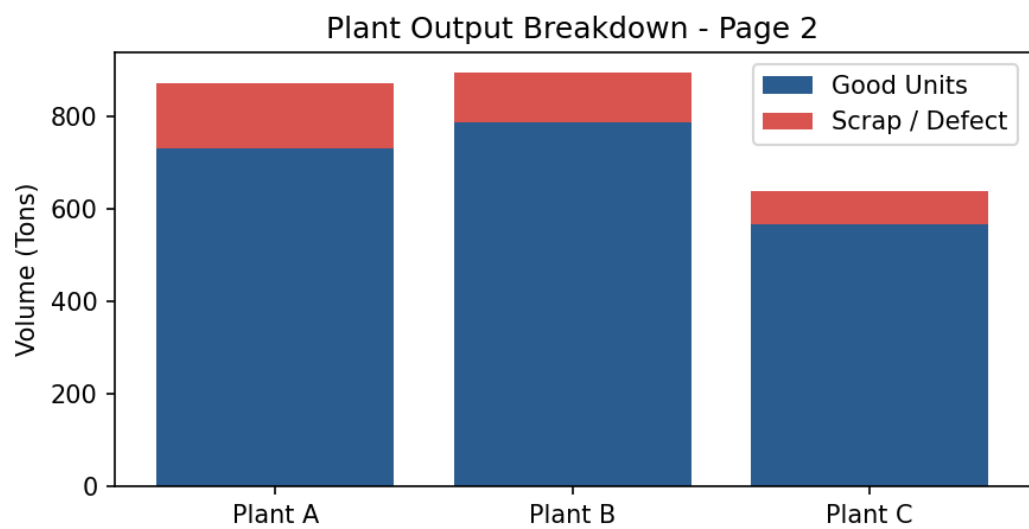
Metric Category	Quarter Baseline	Actual Output*	Efficiency Rate (%)
Line Alpha (Heavy)	1200 Short Tons	1620 Short Tons	92.1%
Line Beta (Light)	800 Short Tons	1134 Short Tons	82.9%
Total Operational Output	2000 Short Tons	2754 Short Tons	87.5%

**Note: Actual output includes unadjusted scrap margin of 3.5% for Short Tons.*

3. Dual-Axis Production vs Efficiency Trend



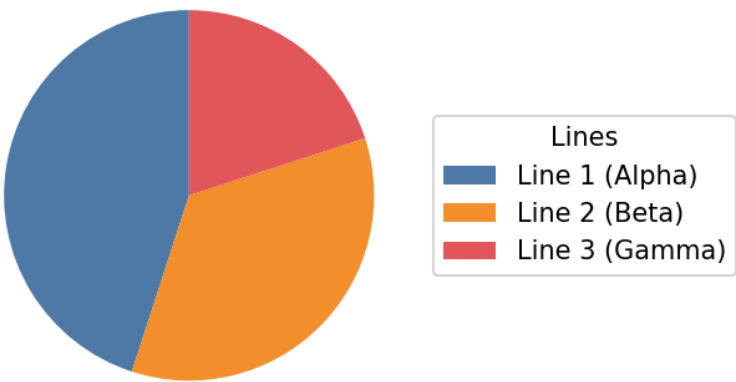
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ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
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C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
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C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

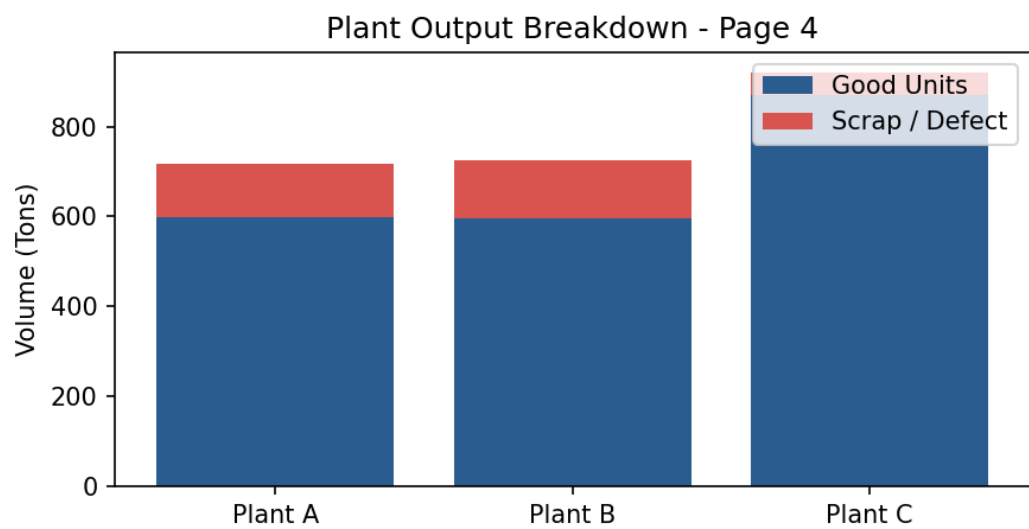
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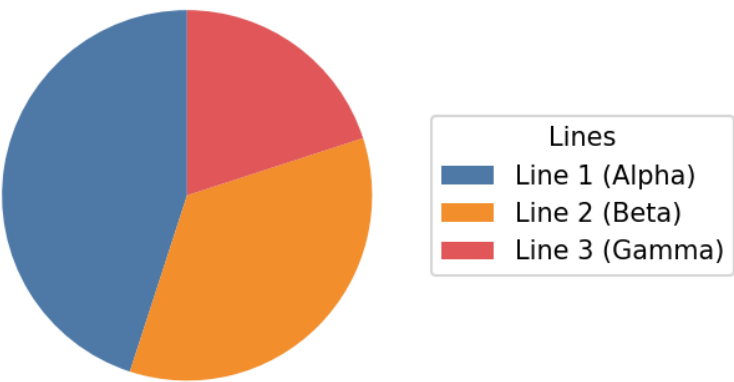
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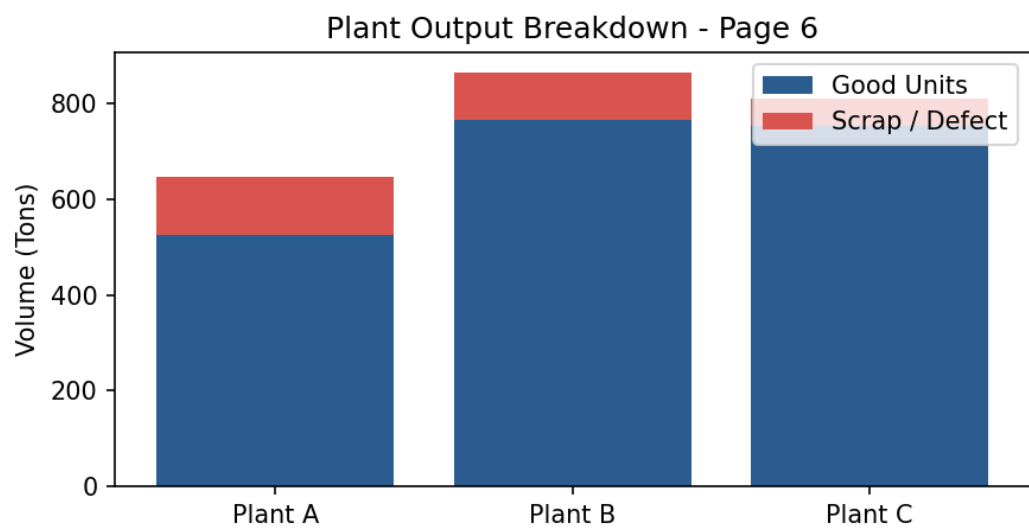
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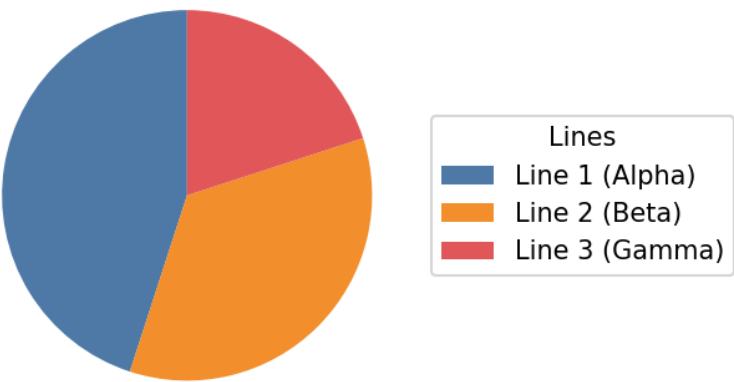
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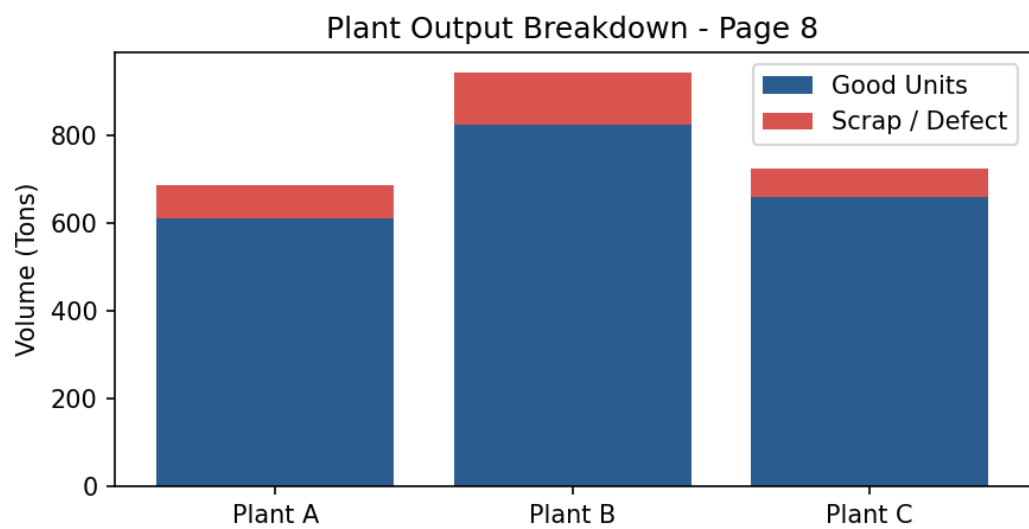
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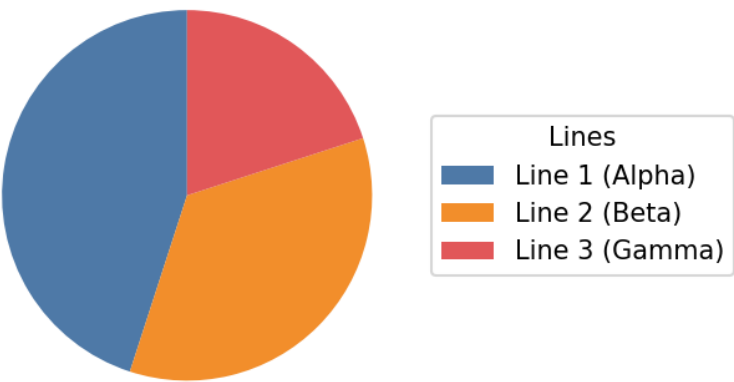
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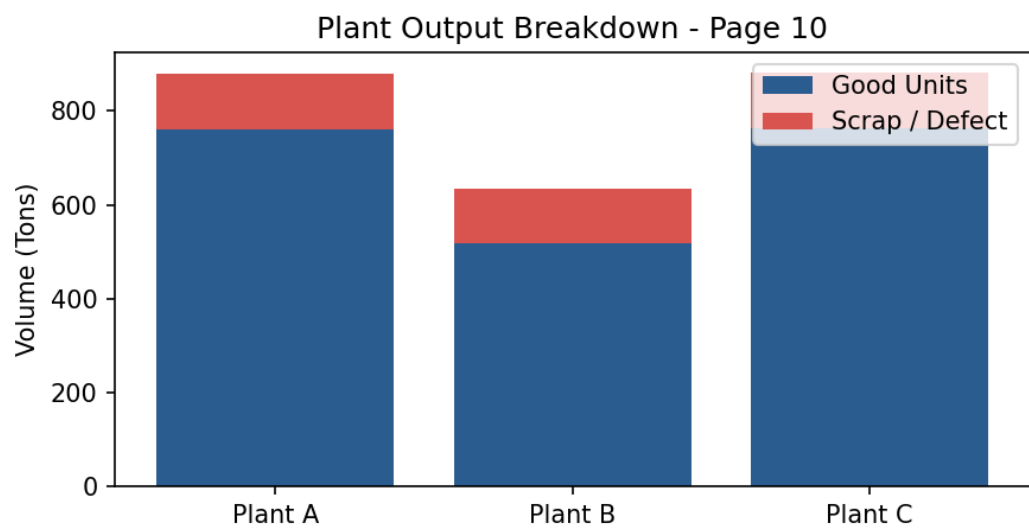
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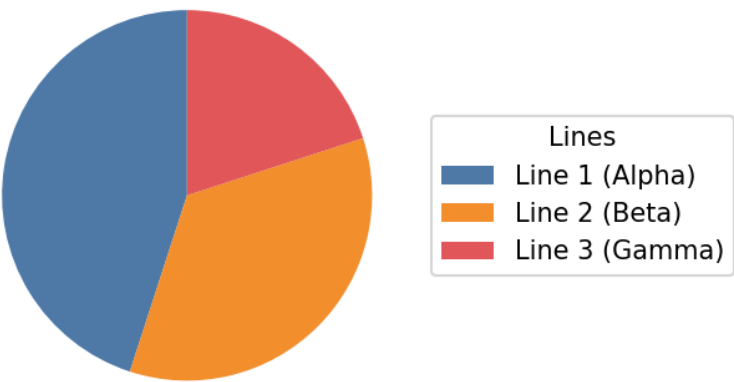
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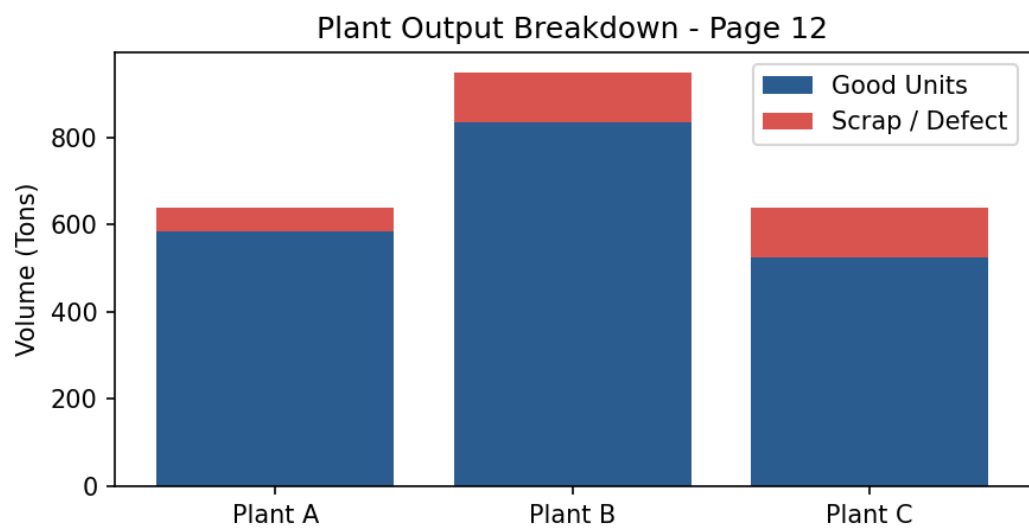
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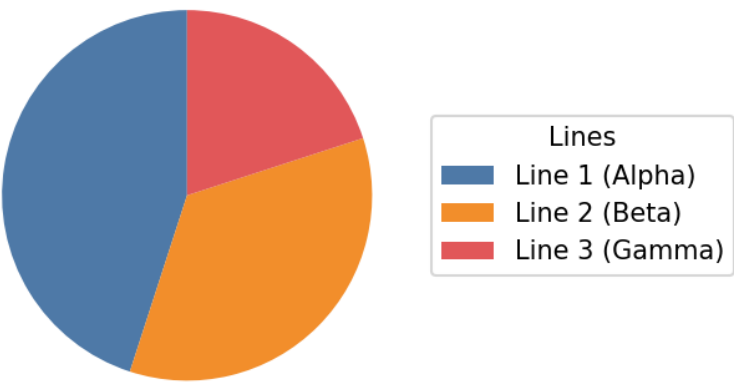
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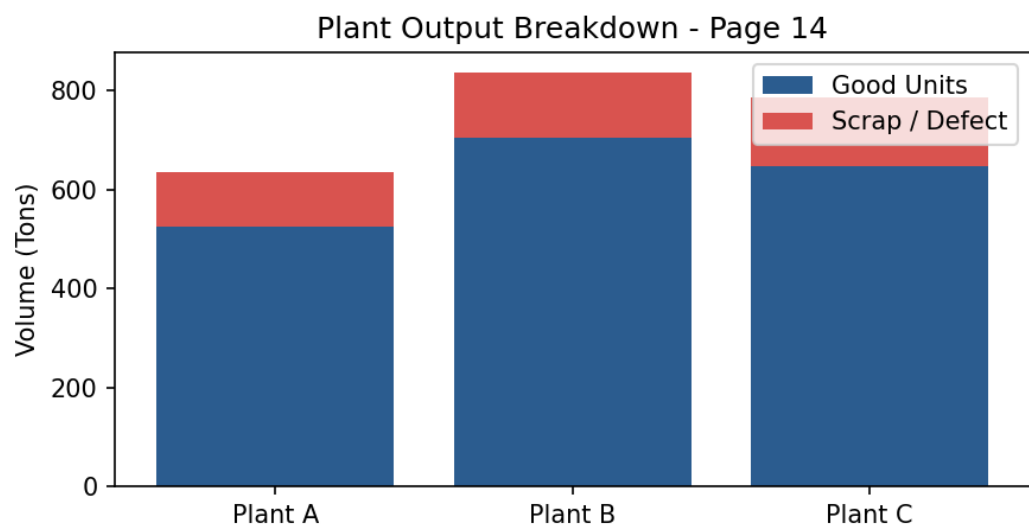
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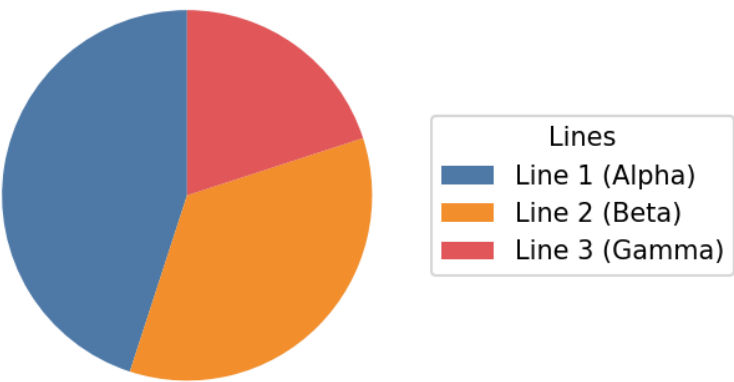
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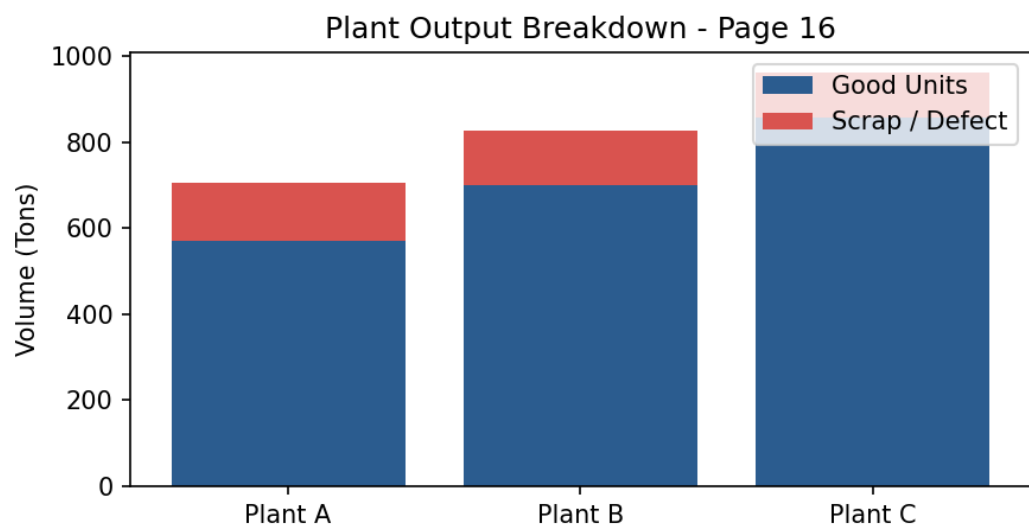
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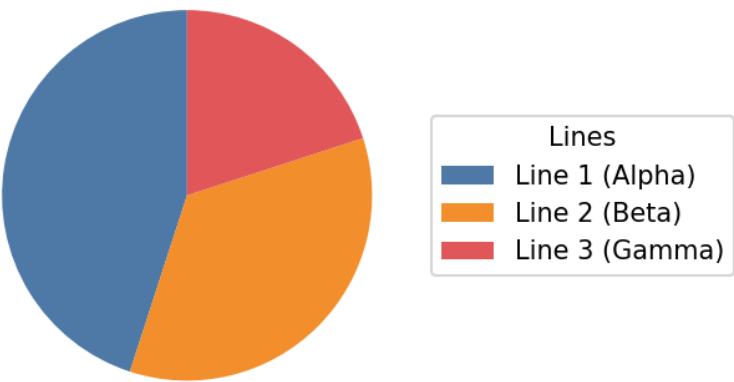
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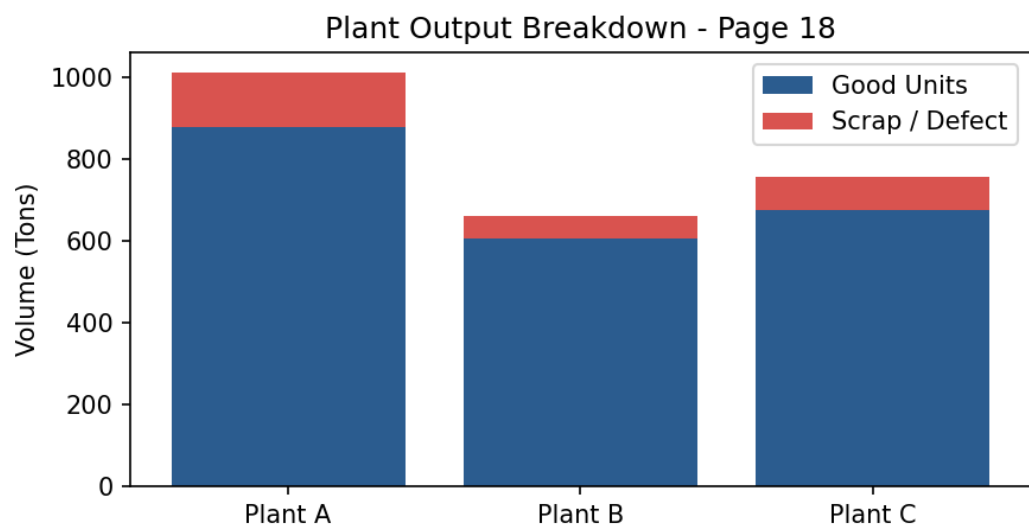
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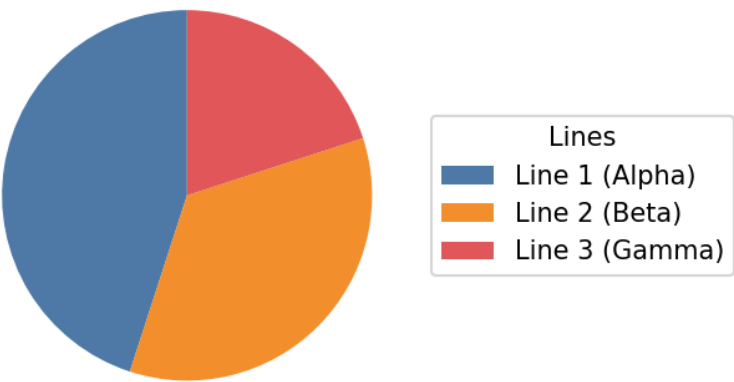
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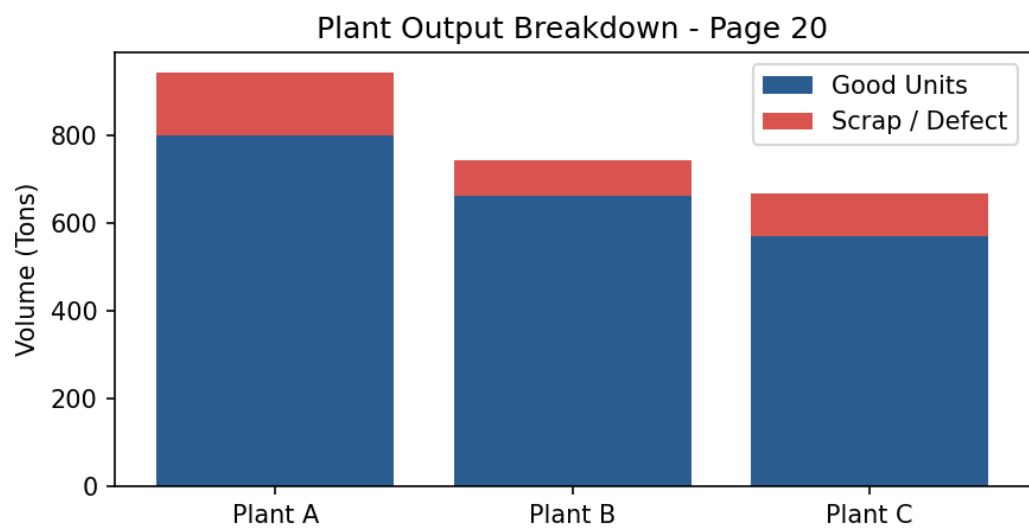
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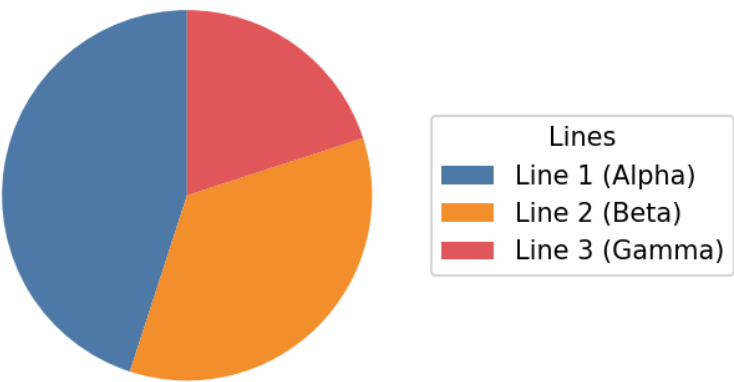
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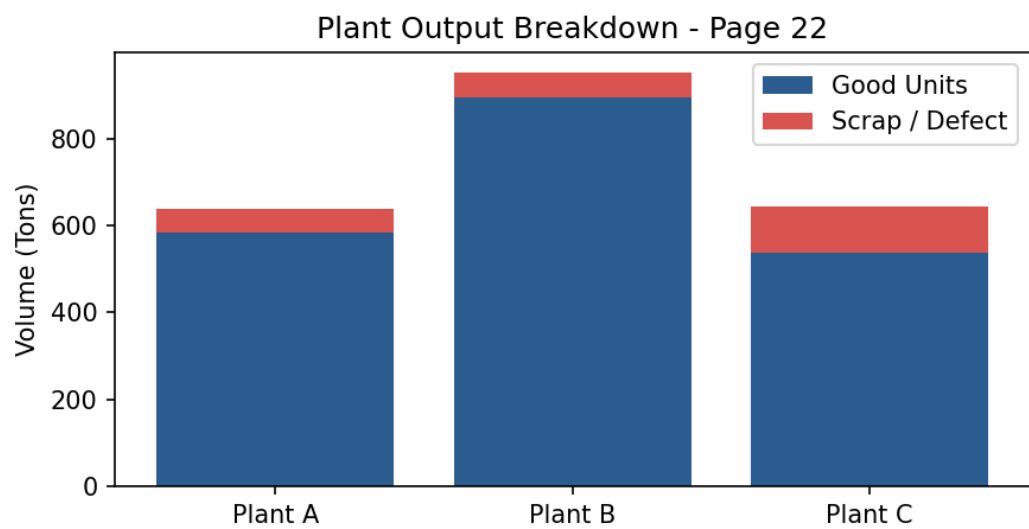
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